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Laser beam oscillation welding of Ta-10W refractory alloy: Molten pool, keyhole behavior and porosity

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Highlights

- Porosity suppressed in Ta-10W non-penetrating LDPW by beam oscillation.
- Molten pool and keyhole behavior linked to Ta-10W materials properties.
- Mechanism of irregular porosity in Ta-10W beam oscillation welds clarified.

Abstract

Ta-10W alloy exhibits excellent high-temperature properties, giving it significant potential for manufacturing critical hot-end components in the aerospace field. To address the issue of porosity defects in laser deep-penetration welding of Ta-10W alloy under atmospheric conditions. This study investigates Laser beam oscillation welding of Ta-10W alloy, analyzing the effects of oscillation radius (r) and frequency (f) on weld formation, porosity, molten pool and keyhole behavior. The results show that when $r \geq 0.3$ mm and $f \geq 150$ Hz, spatter is significantly reduced and porosity is notably suppressed. Under low oscillation parameters, periodic collapse of the rear wall of keyhole leads to the porosity. As oscillation parameters increase, the molten pool and keyhole stability improves when the path overlap ratio exceeds 80%, and beam velocity ranged from 437.90 to 975.82 mm/s (r=0.5, f=150 Hz ~ r=0.5, f=300 Hz), effectively suppressing porosity. The high viscosity and surface tension of Ta-10W alloy results in poor melt fluidity and insufficient remelting due to low-frequency beam oscillation, contributing to slit-shaped and internal-filled porosity. This study clarifies the interaction between oscillation parameters and material properties during laser non-penetrating welding of Ta-10W alloy, providing novel insights for porosity control in laser deep-penetration welding of refractory alloys.

Keywords

Laser beam oscillation welding; Ta-10W; Keyhole behavior; Porosity; "Sandwich" model

1. Introduction

The Ta-10W refractory alloy, with a melting point of 3308K and a service temperature exceeding 1700K, exhibits excellent high-temperature strength, creep resistance, and thermal shock resistance. It demonstrates promising applications in manufacturing critical hot-end components for aerospace and other fields [[1], [2], [3]]. Fiber laser welding, an advanced joining technology characterized by operational flexibility and high process efficiency, has been widely adopted in industrial manufacturing. During atmospheric laser deep-penetration welding (LDPW) of Ta-10W alloy, excessive laser energy density induces instability in the molten pool and keyhole dynamics, triggering spatter and porosity formation, which severely degrade weld quality [[4], [5], [6], [7]]. In 2025, Gong et al. [8] reduced the porosity of Ta-10W welds by vacuum laser welding. However, high cost and geometric constraints limit the industrial application of vacuum welding systems. The porosity defect of Ta-10W alloy has not yet been effectively solved. Using novel processing strategies to stabilize laser welding of Ta-10W and improve weld quality remains critically necessary.

Laser beam oscillation welding (LBOW) has garnered significant research interest recently. Compared to linear laser welding, this method employs controlled oscillatory motion to expand the scanning coverage and flexibility of the laser. Such dynamic modulation enhances process stability during laser welding, effectively suppressing porosity while improving weld quality [[9], [10], [11]]. Horník et al. [12] conducted research and discovered that although the beam energy density distributions of the four common oscillation modes (linear, circular, 8-shaped and infinity-shaped) are similar, the differences in energy distribution along the oscillation trajectories lead to markedly distinct weld formations. The oscillation mode also has a significant impact on laser welding defects. Research shows that the porosity and spatter behavior of linear oscillation welds are always more severe than those of circular oscillation [13,14]. This phenomenon can be explained by the velocity variations along the oscillation trajectory. During linear oscillation, the beam velocity decreases to zero at the amplitude and the direction changes abruptly. This results in energy accumulation at these amplitude positions, thereby increasing the possibility of keyhole eruptions [15]. Conversely, when the oscillation mode is relatively smooth curves such as circular, 8-shaped, or infinity-shaped, the acceleration and velocity of the beam along the oscillatory trajectory vary in a uniform manner and with a relatively small magnitude. Therefore, it is easier to attain better welding quality [[16], [17], [18]]. Among them, circular oscillation has the most uniform beam energy distribution and the smallest velocity variation amplitude and is currently the widely used oscillation mode [19]. For the circular oscillation welding, Franco et al. [16] and Sadeghian et al. [20] on Cu, and Shi et al. [21]on stainless steel welding results show that the severity of spatter and porosity rate are negatively correlated with oscillation parameters (frequency and amplitude), while Zhang et al. [10] converted the circular oscillation parameters into keyhole velocity during welding and found that there is a threshold phenomenon for porosity defects. When the keyhole velocity is greater than 70m/min, the porosity rate can be suppressed under 1%.

The flow of the molten pool and the evolution of keyhole morphology in laser welding are key factors influencing porosity defects. Some studies on circular oscillation welding of aluminum alloys have shown that circular oscillation can stabilize the deep penetration keyhole during welding and generate vortices in the molten pool. This combined effect enlarges the keyhole opening, thereby facilitating pore escape and reducing defect formation [22,23]. Numerical simulations further reveal that circular oscillation stabilizes the keyhole and molten pool and reduces the weld solidification rate, and this delayed solidification allows sufficient time for bubbles to escape from the molten pool before complete solidification [24,25]. And during LBOW, the keyhole moves along the oscillation trajectory. This helps the oscillating keyhole trap bubbles in the molten pool and allow them to escape through the keyhole. As a result, weld porosity is significantly reduced [23,26]. In addition, some studies on the LBOW of aluminum and steel have indicated that even when pores are present after welding, most of these pores are regularly shaped as circular or oval shaped [14,16]. In some cases, they may appear as elongated oval or triangles shaped, with their interiors unfilled [17]. The pore morphology is comparable to that of no-oscillation welding, showing no remarkable discrepancies.

Circular beam oscillation effectively suppresses porosity formation in laser welding. This effect has been confirmed in multiple metallic systems, such as aluminum alloys, titanium alloys, steel and superalloys [15,21,22,27]. However, most scholars have mainly discussed the relationship between process parameters and weld quality as well as molten pool fluctuations, without associating the behavior of the molten pool and keyhole during the LOBW process with the properties of metal melt. And research on LBOW of Ta-10W alloy remains unexplored. Additionally, the alloy exhibits a high density (16.8 g/cm^3) and an elevated melting point (3308 K), and the influence of its unique material properties on molten pool/keyhole dynamic behavior, and porosity formation mechanisms has not been systematically explained. To address this issue, this study employs high-speed camera synchronous observation methodology to systematically investigate the characteristics of weld morphology, molten pool and keyhole dynamics behavior under varying oscillation parameters. By correlating the alloy material properties with process behaviors, we elucidate the mechanisms governing weld formation and porosity evolution during LBOW of Ta-10W alloy.

2. Materials and methods

2.1. Materials

The experimental Ta-10W alloy plates, provided by CNMC Co., Ltd., were prepared via electron beam melting, followed by rolling and full annealing. The material properties of the alloy are summarized in Table 1. Two plate dimensions were used: 60 mm×20 mm×4 mm for welding process optimization trials and 70 mm×20 mm×5 mm for “sandwich” model welding. All specimens were surface-ground and ultrasonically cleaned in acetone before welding.

Table 1. Material properties of Ta-10W alloy [28].

Material	**Ta-10W alloy**
Melting point	3308 K
Density	16.8 g·cm^{-3}
Thermal conductivity	46 W·m^{-1}·K^{-1}
Thermal expansion	6.7×10^{-6} K^{-1}

2.2. Laser beam oscillation welding

Fig. 1 illustrates the experimental setup for LBOW. The system employs an IPG YLS-6000 multimode fiber laser with a wavelength of 1060–1070 nm and a 200 μm core diameter transmission fiber. Beam oscillation is achieved using a Sino-Galvo HP30–6000 high-power galvanometer scanner, a 120 mm collimation lens, and a 170 mm F-θ lens. A trailing-shield welding nozzle (Fig. 1 a, b) provided inert gas protection of the molten pool and high-temperature weld zone. The nozzle maintained a 1 mm standoff distance from the plate surface, and the opening size of the nozzle tail was 30 mm×4 mm (length × width) [29]. Constant laser parameters were maintained: P=4 kW, Δf=0 mm. High-purity argon (99.999%) shielding gas was supplied at a 12 L/min flow rate.

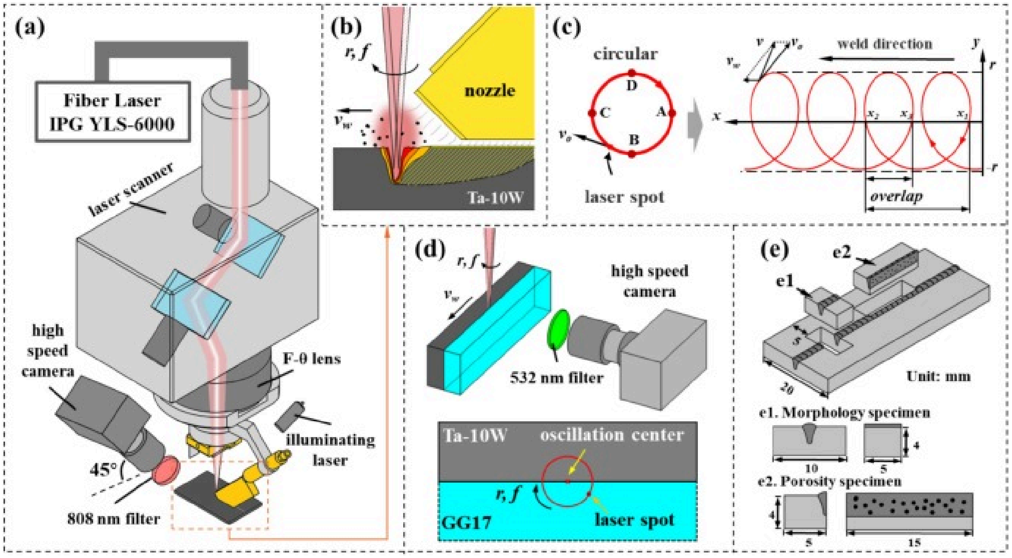

Fig. 1. Experimental setup of LBOW of Ta-10W alloy and analyses: (a) overview; (b) enlarged view of molten pool; (c) oscillation pattern and path; (d) schematic diagram of "sandwich" model welding; (e) sampling methods of specimens for morphology and porosity examinations.

As shown in Fig. 1 c, the experiment employed a circular oscillation mode, with points A, B, C, and D on the circular path in Fig. 1 c representing four characteristic positions. The welding path was controlled using a linear displacement platform and galvanometer scanner. By superimposing beam circular oscillation onto linear motion, the beam path during oscillatory welding can be described as compound motions in the *x* and *y* directions as follows:

$$\begin{cases} x_t = v_w t + r cos(2\pi f t + \varphi) \\ y_t = r sin(2\pi f t + \varphi) \end{cases} \quad (1)$$

where v_w is the linear speed (maintained at 2 m/min during experiments), *r* is the oscillation radius (mm), *f* is the oscillation frequency (Hz), φ=−π/2 is the initial phase, and *t* is the welding time (s).

The beam oscillation velocity can be derived as:

$$\begin{cases} v_o = 2\pi r f \\ v_x(t) = v_w - v_o\, sin(2\pi f t + \varphi) \\ v_y(t) = v_o\, cos(2\pi f t + \varphi) \\ v_t = \sqrt{{v_x}^2(t) + {v_y}^2(t)} \end{cases} \quad (2)$$

where v_o and v_t are the beam circular oscillation and actual velocities, respectively; $v_x(t)$ and $v_y(t)$ are velocity components in *the* *x* and *y* directions, respectively.

Furthermore, the beam circular oscillation induces path overlap, which ratio is defined as [30]:

$$Overlap\ ratio = \frac{x_2 - x_3}{x_2 - x_1} \times 100\% \quad (3)$$

where x_1, x_2, and x_3 represent the x-coordinates of successive intersection points between the oscillating beam path and the *X*-axis during a single cycle, as illustrated in Fig. 1 c.

This study primarily analyzed the weld under 12 oscillation parameters listed in Table 2 and compared these with the welding tests conducted without oscillation (designated as 0#).

Table 2. Experimental oscillation parameters.

		Oscillating frequency f / Hz		
		50	**150**	**300**
Oscillating amplitude r / mm	**0.1**	1#	2#	3#
	0.2	4#	5#	6#
	0.3	7#	8#	9#
	0.5	10#	11#	12#

2.3. Characterization and analysis methods

After the welding process, the surface morphology of the samples was observed. Cross-sectional morphology specimens (Fig. 1 e1) and longitudinal porosity specimens (Fig. 1 e2) were extracted from each weld. The specimen dimensional specifications and sampling locations are indicated in Fig. 1 e. The cross-sectional morphology specimens were prepared by mechanical polishing and etching, followed by morphological observation using a Keyence VK-X130K optical microscope. The volume ratio of the etchant was H_2SO_4: HNO_3: HF=5:2:4 and the etching time was approximately 15 s [31]. Longitudinal porosity specimens were sectioned along the weld center plane and analyzed using ImageJ software for the quantification of pores area distribution. To preserve pore boundary integrity in the welds, these specimens were mechanically polished without etching before image analysis, enabling precise correlation between oscillation parameters and porosity characteristics.

A high-speed camera (Photron FASTCAM Mini UX100) equipped with a Nikon AF lens was synchronized to observe the dynamic behavior of the molten pool during welding, as shown in Fig. 1 a. The camera was tilted at a 45° angle, and an infrared semiconductor illumination laser was placed on the opposite side. The camera operated at a frame rate of 10000 fps. To ensure optimal imaging quality, a narrow-band filter (808 nm transmission wavelength) and an attenuation filter (20% transmittance) were installed in front of the lens during recording.

On this basis, welding experiments using a “sandwich” model were conducted to characterize the keyhole behavior during LBOW of Ta-10W alloy through high-speed imaging. The “sandwich” model experiments aimed to elucidate the mechanisms of pore generation and suppression under beam oscillation conditions. The schematic of the “sandwich” model is shown in Fig. 1 d. The model comprised GG17 glass and Ta-10W plates in a butt joint configuration. The mating surfaces (metal-glass interface) were precision-ground and securely clamped with customized fixtures to ensure gas-tightness. During welding, the center of the circular oscillation path (corresponding to the laser-focal point in linear welding) was precisely aligned with the metal-glass interface. The high-speed camera was horizontally positioned on the glass side to capture keyhole dynamic behavior at the metal-glass interface through the glass. For the “sandwich” model, high-speed imaging employed a 532 nm narrow-band filter without an illumination laser. At the same time, other parameters remained consistent with those for high-speed imaging of the molten pool.

3. Experimental results

3.1. Weld morphology of LBOW

The surface and cross-sectional morphologies of welds under different parameters (specimens 0#~12#) are shown in Fig. 2. All welds exhibit good formation and effective protection, characterized by bright metallic luster without oxidation or cracking. With increasing radius and frequency, significant changes were observed in cross-sectional morphology: undercut and spatter of the welds were reduced, and surface formation quality improved. In addition, three characteristic dimensions, namely weld surface width (W_s), weld penetration depth (D_w), and weld width at 1 mm depth (W_1), were selected to analyze the formation regular of LBOW in Fig. 2 quantitatively.

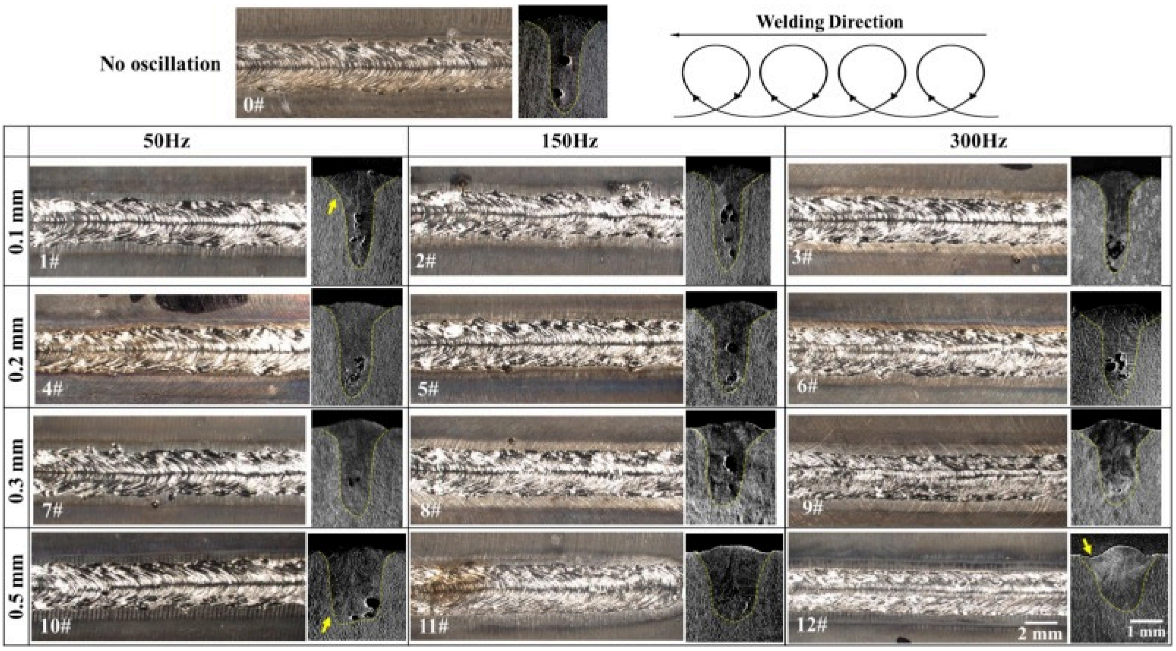

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Fig. 2. Weld surface and cross-sectional morphologies of LBOW.

The corresponding measurement results are presented in Fig. 3. Where W_s=2.239 mm under no oscillation welding conditions (0#). In this study, the maximum beam oscillation radius was 0.5 mm, corresponding to a scanning range smaller than half of the W_s value of the weld 0#. Consequently, the W_s under all oscillation parameters remained in the range of 2.2–2.5 mm, exhibiting comparable dimensions to the weld 0#.

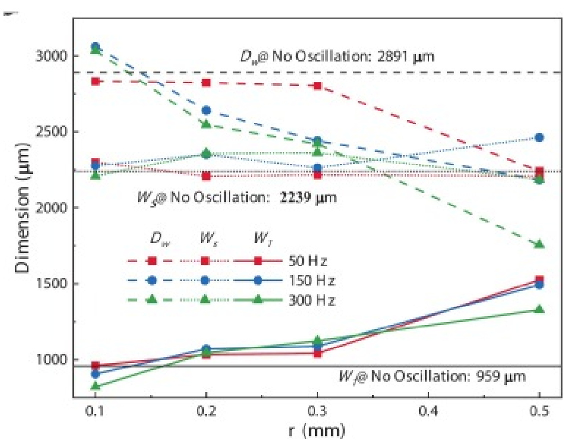

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Fig. 3. The statistical results of LBOW weld cross-sections D_W, W_1 and W_s.

As shown in Fig. 2 and Fig. 3, the oscillation parameters significantly influence weld cross-sectional morphology. At an oscillation frequency of 50 Hz, pronounced asymmetry was observed at the weld bottom (indicated by yellow arrows). This asymmetric characteristic diminished with increasing frequency. When the frequency reaches 300 Hz, a slight undercut appears on the cross-section of weld 12#. D_W is generally negatively correlated with radius and frequency: when r=0.1–0.3 mm, the welds exhibited nail-head morphologies; at r=0.5 mm, D_W significantly decreases, and the morphology changes from nail-head to columnar at 50 Hz and 150 Hz; when the frequency increases to 300 Hz, D_W is only 1.75 mm, presenting a bowl shape, and the weld morphology transitions between deep penetration and thermal conduction. Notably, for r=0.1 mm and f=150–300 Hz (welds 2# and 3#), the enhanced remelting effect caused by high-frequency beam oscillation results in D_W slightly exceeding that with no oscillation welds. In LDPW, the laser acts on the front wall of the keyhole [5]. During LBOW, the beam scanning range continuously increases with the increase in radius, and the front wall of the keyhole expands to both sides under the drive of the laser beam, resulting in a positive correlation between W_1 and radius.

This phenomenon is correlated with the beam oscillation-induced modulation of the laser energy distribution during LBOW. The spatial power density profile of the laser exhibits an approximately Gaussian distribution, which can be described as follows:

$$I(X,Y) = I_0 \bullet exp\left(\frac{-2X^2}{{R_0}^2}\right) \bullet exp\left(\frac{-2Y^2}{{R_0}^2}\right) \quad (4)$$

where I_0 is the maximum power density at the center of the laser spot. For a laser beam with a spot radius of R_0, I_0 solely depends on the laser power.

This study kept parameters including laser power and defocus distance constant. Thus, the maximum energy density of the beam during no oscillation welding was normalized to 100% to evaluate the impact of oscillation parameters on laser energy distribution. Based on Eqs. (1), (4), the energy distributions and laser beam paths under specific oscillation parameters were obtained, as shown in Fig. 4 a-f.

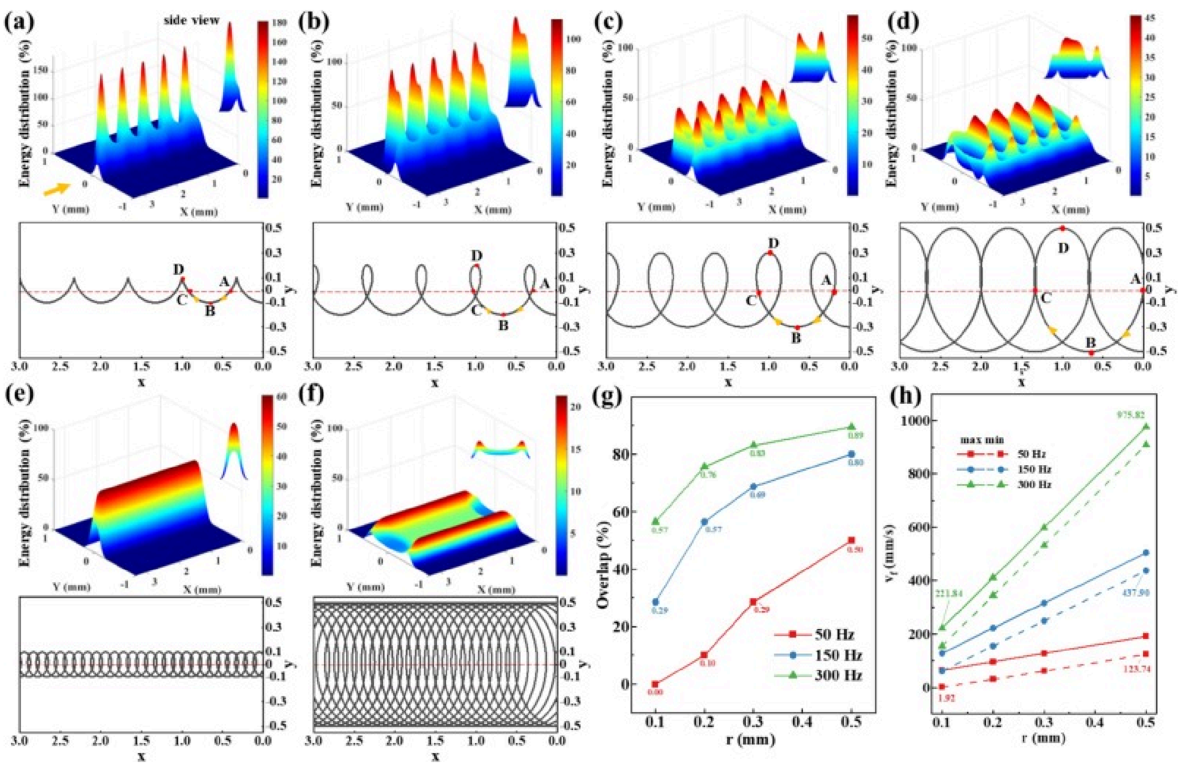

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Fig. 4. Energy distribution, laser beam path, overlap ratio and velocity of LBOW: (a) 1#: r=0.1 mm, f=50 Hz; (b) 4#: r=0.2 mm, f=50 Hz; (c)7#: r=0.3 mm, f=50 Hz; (d) 10#: r=0.5 mm, f=50 Hz; (e) 3#: r=0.1 mm, f=300 Hz; (f) 12#: r=0.5 mm, f=300 Hz; (g) the overlap ratio of oscillation path; (h) max and min of oscillating laser beam.

The side view of each energy distribution is provided in the upper-right of the subfigure. Additionally, the path overlap ratio and the extreme value of beam velocity were calculated for all parameters (Fig. 4 g, h). The results indicate that the overlap ratio and the beam velocity positively correlate with frequency and radius.

With the increased radius, the energy peak of the oscillation laser beam exhibits a decreasing trend. At 50 Hz frequency, the path overlap ratio remained low, leading to localized energy concentration along one side of the oscillation path. Under parameter 1# (Fig. 4 a), the beam path exhibited complete non-overlapping, where point D showed the highest energy density and exceeded the no oscillation condition. This is because at this parameter, the beam velocity at point D is the slowest, merely 1.92 mm/s, which is significantly lower than the 2 m/min (≈33.33 mm/s) during linear welding. When the radius increased to 0.5 mm, the path partially overlapped, and the beam velocity increased. While this effectively reduced the issue of excessive local energy density, the energy distribution remained uneven. Despite the path overlap at points A and C when r=0.5 mm, there was a low-energy zone in the central region of the scanning area, as shown in Fig. 4 d. The path overlap ratio increased with increasing oscillation frequency, and the beam velocity significantly accelerated. This led to gradual homogenization of energy distribution, ultimately forming the symmetrical U-shaped energy distribution.

For LBOW welds, asymmetric cross-sectional formation results from the uneven energy distribution of the oscillation beam at 50 Hz [32]. When employing radii of 0.1 mm (1#), 0.2 mm (4#), and 0.5 mm (10#) (Fig. 4 a, b, d), adjacent oscillation cycles exhibit low path overlap. The beam velocity reaches its maximum at point B and minimum at point D, resulting in maximum energy density at point D. This promotes sufficient beam-material interaction and more melting at D while minimal melting at B. At r=0.1 mm, the bottom of the molten pool is narrow, and the melting volume is mainly reflected in the curvature of the fusion line. Thus, at the nail-head position of weld 1#, the fusion line on the left side is gentle, while there is a distinct bend on the right side. Larger radii (4#, 10#) induce asymmetrical energy peaks between weld sides, leading to slightly greater penetration depth on the left side (welds 4# and 10# in Fig. 2). However, at r=0.3 mm (weld 7# in Fig. 2 and Fig. 4 c), despite comparable energy peaks on both sides, repeated thermal input at path crossover points leads to deeper penetration on the right side [30].

3.2. Pore morphology and statistics of LBOW

Porosity is one of the most prevalent defects in LDPW. As observed in the cross-sectional morphologies of the LBOW in Fig. 2, severe porosity occurs under some vibration parameters. To comprehensively investigate the relationship between oscillation parameters and porosity, longitudinal sections along the weld center plane were observed. Fig. 5 presents porosity morphologies from these sections across all welds, with pore boundaries outlined in yellow for clarity. Fig. 5 shows that, except for welds 9#, 11#, and 12#, significant porosity is observed in other welds. In welds 0# to 8#, both large and small pores are obvious, whereas weld 10# exhibits primarily large pores. The weld porosity exhibits typical keyhole-induced porosity characteristics, primarily distributed in the central and lower regions of the welds. The larger pores in the upper region of the weld originate from smaller pores, which floated upward and grew before solidification.

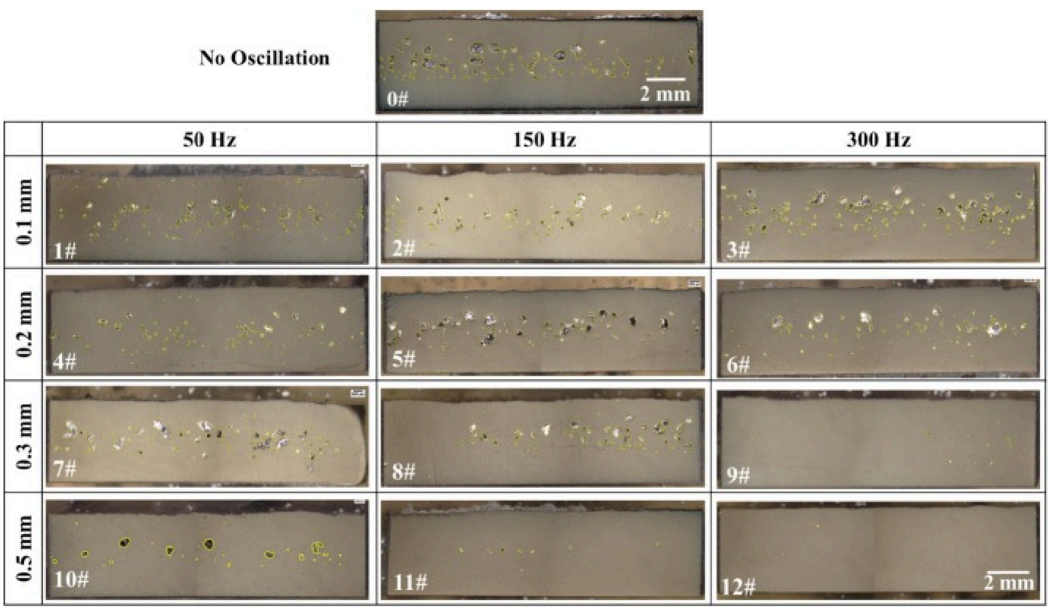

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Fig. 5. Weld pore morphologies of LBOW.

Based on the results in Fig. 5, ImageJ software was used to identify and statistically analyze the quantity and area of pores, with quantitative results shown in Fig. 6. The horizontal reference lines in Fig. 6 a and Fig. 6 b indicate the corresponding results obtained under no oscillation welding conditions. Fig. 6 a show the variation in the pore number and the porosity with respect to the oscillation parameters. The calculation method of porosity is as follows:

$$Porosity = \frac{S_p}{S_w} = \frac{S_p}{D_w \times 15000\ \mu m} \tag{5}$$

where S_p is the total area of pores, S_W is the longitudinal sections area of welds, D_W is the weld penetration depth as shown in Fig. 3, 15000μm is the length of the porosity specimen. The pore number is related to the oscillation parameters (as shown in Fig. 6 a). At f=50Hz and 150Hz, the pore number initially increased and then decreased with an increasing oscillation radius. In contrast, at 300Hz, a negative correlation between pore number and oscillation radius was observed. Except for r=0.1mm, the pore number exhibits an inverse relationship with frequency. Furthermore, the porosity of each weld, as shown on the right axis of Fig. 6 a, is compared with that of the no oscillation welding. The porosity levels in all welds, with the exception of weld 7#, are lower compared to those observed in welds produced without oscillation. When r≤0.2mm, no significant correlation is observed between changes in porosity and the oscillation parameters. However, at r=0.3mm and 0.5mm, porosity exhibits a negative correlation with f. In welds 9#, 11#, and 12#, the porosity are rated at 0.39%, 0.20%, and 0.12%, respectively. The porosity is suppressed to an exceptionally low level and is nearly negligible. Fig. 6 b displays the variation in maximum and average pore areas with oscillation parameters. For r=0.1mm and 0.2mm, the maximum pore area demonstrates a positive correlation with frequency. In contrast, at r=0.3mm and 0.5mm, an inverse relationship is observed between maximum pore area and frequency. The average pore area follows a similar pattern to the maximum pore area.

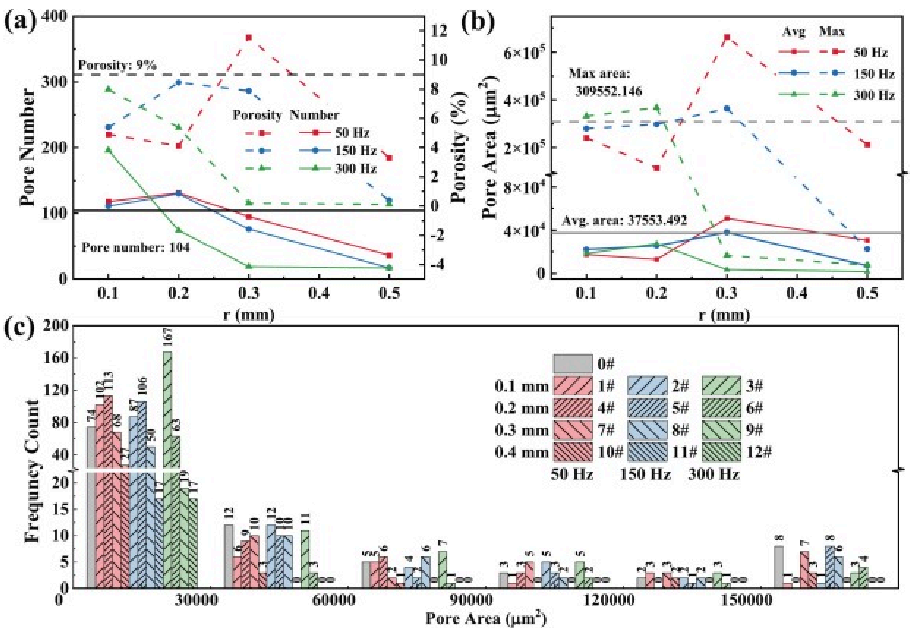

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Fig. 6. Weld pores statistics of LBOW: (a) pore number and porosity; (b) pore average and maximum area; (c) frequency distribution of pore area.

For LBOW welds with r≥0.3mm, the pore number is consistently lower than that of the non-oscillation weld 0#. Similarly, weld 7# is an exception: its maximum and average pore areas exceed those of weld 0#. Fig. 6 c shows the frequency distribution of the pore area. For all oscillation parameters, the porosity of the weld is mainly small pores with S<30000μm^2, accounting for over 70% of the total porosity. The variation in the quantity of these small pores with oscillation parameters is consistent with the total pore number. In welds 9#, 11#, and 12#, only a few small pores are observed, with no pores area exceeding 30000μm^2. Under these parameters, the porosity was significantly reduced when the beam velocity exceeded 437.90mm/s. Additionally, the pore number exhibits a correlation with cross-sectional morphologies. While W_s values are similar for all welds, increasing r and f enhances path overlap ratio and expands the beam scanning area. Therefore, the widened W_1 promotes bubble growth, rise, and eventually escape from the molten pool. Notably, the welds 9#, 11#, and 12# achieved path overlap ratio of 80%, 83%, and 89%, respectively. When the path overlap ratio exceeds 80%, and the beam velocity exceeds 437.90mm/s, porosity formation is effectively suppressed in laser welds of Ta-10W alloy.

3.3. Molten pool and keyhole dynamic behavior

Applying circular beam oscillation in Ta-10W laser welding changes the porosity formation rule compared to the no-oscillation welding condition. In laser welding, porosity defects are related to the dynamic behaviors of molten pool and keyhole [33]. Based on the porosity statistics in Section 3.2, two groups of welds were selected: Group 1 (welds 1#, 3#, 10#) with pronounced porosity, and Group 2 (welds 9#, 11#, 12#) with effectively suppressed porosity. The dynamic behavior of the molten pool and keyhole within a single oscillation cycle under different oscillation parameters were investigated and compared with the no oscillation weld (0#). The mechanisms of porosity formation in Ta-10W laser welds were further discussed.

Fig. 7 shows the dynamic behavior of the molten pool and keyhole for no-oscillation welding (0#). Under this parameter, the molten pool and keyhole behavior do not show obvious periodic characteristics. Under the Marangoni effect, the molten metal in the molten pool flows from both sides of the keyhole toward the rear region and gradually solidifies (as shown in Fig. 7 a). During welding, intense metal vapor eruptions occur in the molten pool and keyhole, forming molten droplets and spatters around the molten pool (indicated by the yellow arrows). The formation of these molten droplets and spatters exhibits no obvious regularity. Fig. 7 b presents the welding results of the "sandwich" model with parameter 0#. The bright green region observed at the metal-glass interface corresponds to the keyhole and surrounding molten metal. The keyhole is filled with metal vapor. The bright white spot indicates the instantaneous position of the laser, which exhibits periodic vertical oscillation along the keyhole front wall with a cycle duration of approximately 0.5–0.9 ms. At t_0, the bright spot is located at the lower of the keyhole, while the keyhole rear wall exhibits irregular protrusions induced by vapor recoil pressure effects. At t_0+0.4 ms, the bright spot disappeared and the width at the lower of the keyhole decreased. At t_0+0.7 ms, an obvious "necking" was observed at the upper of the keyhole and it weakened again at t_0+0.9 ms. These findings confirm that the periodic keyhole fluctuations and the collapse of rear wall molten metal in no oscillation welding are the primary causes of pores formation.

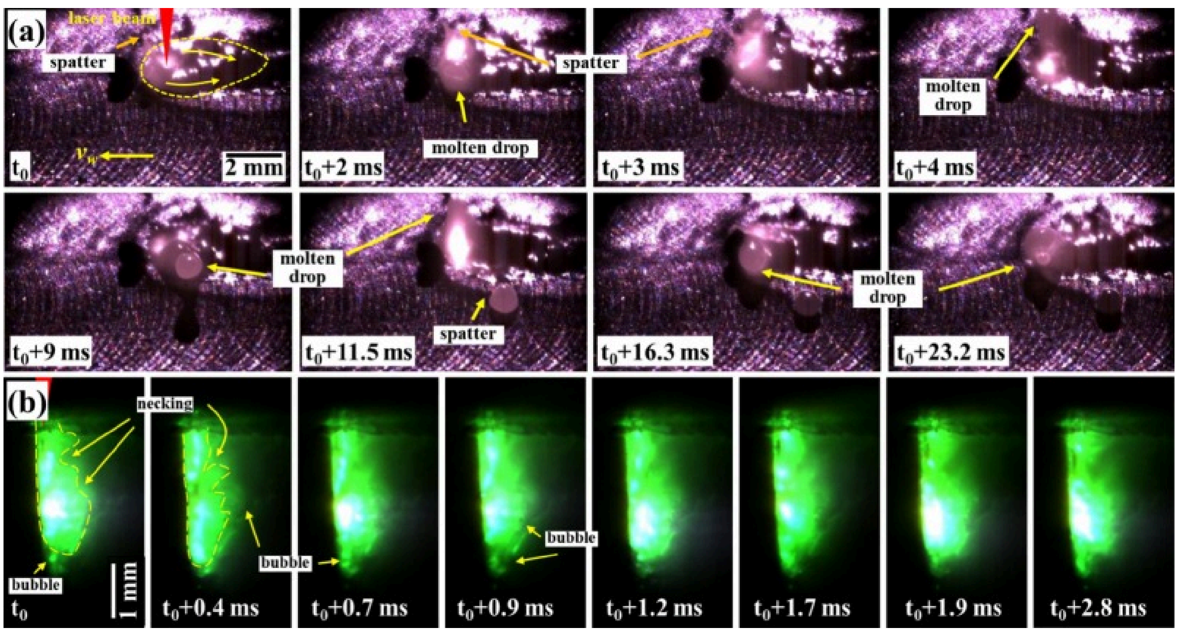

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Fig. 7. Molten pool and the keyhole behavior of no oscillating laser welding: (a) molten pool; (b) keyhole.

In addition, high-speed images of the molten pool for all parameters were used to measure its length, as shown in Fig. 8 a. The images of the "sandwich" model were utilized to measure fluctuations in keyhole opening dimensions at the metal-glass interface over a 300 ms period for parameters 0#, 1#, 3#, and 9# to 12#, as shown in Fig. 8 b, c. The molten pool length positively correlates with the oscillation radius at all frequencies. The application of oscillation increases in the beam velocity (as shown in Fig. 4 h). Therefore, when the radius is 0.1–0.2 mm, the length of the molten pool is less than that of the no oscillation welding.

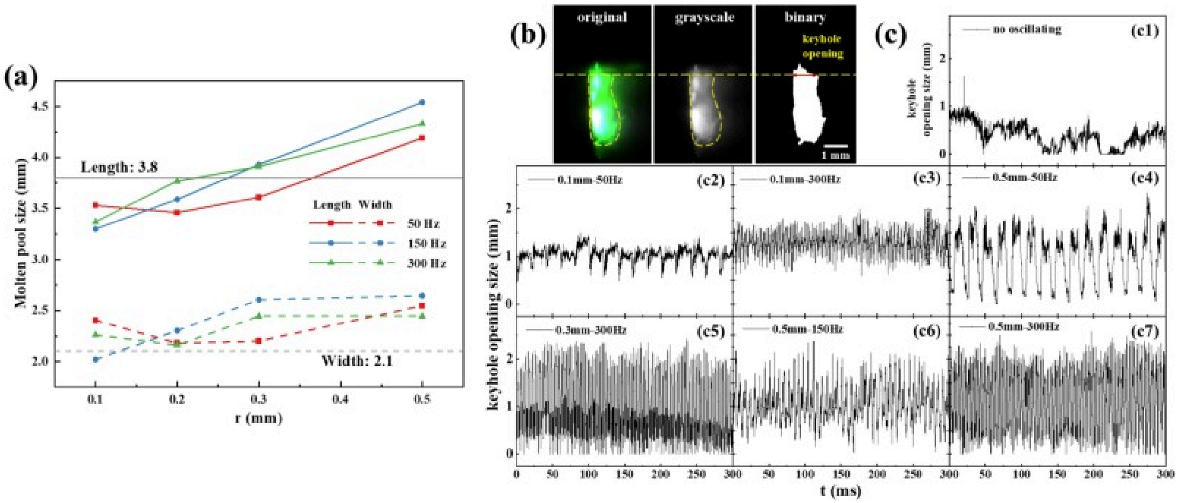

Fig. 8. Molten pool and keyhole size during welding: (a) molten pool length; (b) schematic diagram of keyhole opening size; (c) curve of keyhole opening size and time of 0#, 1#, 3#, 9#, 10#, 11#, and 12#.

The method for acquiring keyhole opening size is shown in Fig. 8 b: Image sequences from the “sandwich” model undergo grayscale conversion and binary processing, followed by the white pixel width extraction at the keyhole top. This width represents the keyhole opening size at the metal-glass interface, with results in Fig. 8 c. For no oscillation welding, keyhole opening size is primarily distributed below 1 mm and the fluctuation is unstable. The applied oscillation drives the laser beam to reciprocate across the metal-glass interface for parameters 1#, 3#, 9#, 10#, 11#, and 12#. This reciprocating motion increased the maximum opening size of the keyhole at the interface, expanded the fluctuation amplitude, and induced periodic change in the keyhole opening. At r=0.1 mm, the maximum keyhole opening size is approximately 1.5 mm, and the fluctuation range is about 1 mm; while when radii increase to 0.3 mm and 0.5 mm, the maximum opening size and fluctuation range at the interface are about 2.5 mm. Higher frequency is also conducive to maintaining a larger keyhole opening size at more times. The combined effects of an elongated molten pool and enlarged keyhole opening size promote bubble escape during welding, thereby suppressing the porosity defect.

Fig. 9 shows the welding molten pool dynamic behavior under some oscillation parameters. Points A, B, C, and D correspond to different positions on the oscillation path, as defined in Fig. 1 c. High-speed imaging analysis shows that in LBOW of Ta-10W, spatter is mainly found at small radii and low frequencies, and decreases with the increase in oscillation radius or frequency (as shown in Fig. 9 a-c). At small radii (0.1 mm), the limited oscillation range restricts keyhole size; in this condition, the higher metallic vapor pressure in the keyhole induces a spatter. The oscillation path exhibits a lower overlap ratio for the r=0.5 mm and f=50 Hz condition (Fig. 4 a, d). When the beam approaches point C, intense energy coupling between the laser and solid metal generates depression at the molten pool edge and causes a spatter (Fig. 9 c3-c4).

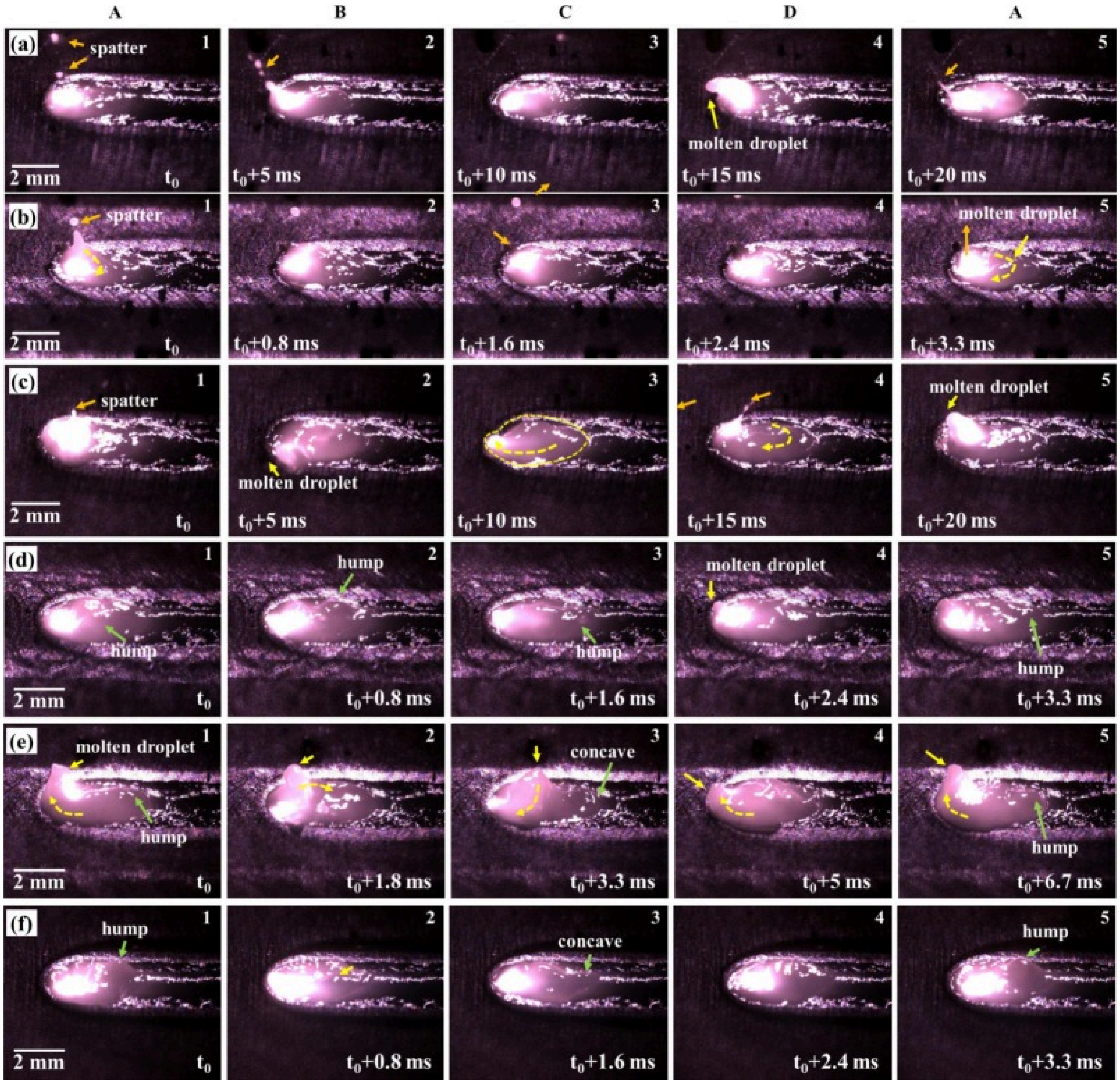

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Fig. 9. Molten pool behavior of LBOW welds: (a) 1#: r=0.1 mm, f=50 Hz; (b) 3#: r=0.1 mm, f=300 Hz; (c) 10#: r=0.5 mm, f=50 Hz; (d) 9#: r=0.3 mm, f=300 Hz; (e) 11#: r=0.5 mm, f=150 Hz; (f) 12#: r=0.5 mm, f=300 Hz.

Fig. 10 shows the schematic of metal flow in the molten pool during linear welding and LBOW. During linear welding, longitudinal metal flow in the molten pool is mainly driven by recoil pressure and thermal gradients. During LBOW, circular beam oscillation induces horizontal vortex in the molten pool, and the vortex intensity positively correlates to beam velocity [34]. At low frequencies, molten droplets form significantly above the molten pool (Fig. 9 a4, b1, c5), which is mainly attributed to reduced beam velocities under low frequencies. The lower vortex intensity in the molten pool allows recoil pressure to dominate, driving longitudinal melt flow toward the surface and resulting in molten droplets. For parameter 1#, at point D, the laser beam has the lowest velocity, which allows for sufficient interaction between the laser and metal, intensifying the metallic vapor pressure in the keyhole and generating large molten droplets ahead of the molten pool. When protruding molten metal at relatively high velocities, it becomes easy to detach and form spatters (as shown in Fig. 9 b1). At large radii and high frequencies, the increase in the keyhole opening size makes it easier for metallic vapor to escape. Concurrently, accelerated beam intensifies the vortex in the molten pool. When the effect of the horizontal vortex on the molten metal is stronger than the longitudinal recoil flow, spatter and molten metal humps are suppressed. This results in the molten pool becoming more stable. The molten metal bulges and falls along the edge of the molten pool in a clockwise direction under the laser beam driven. The fluctuation cycle is basically consistent with the oscillation cycle.

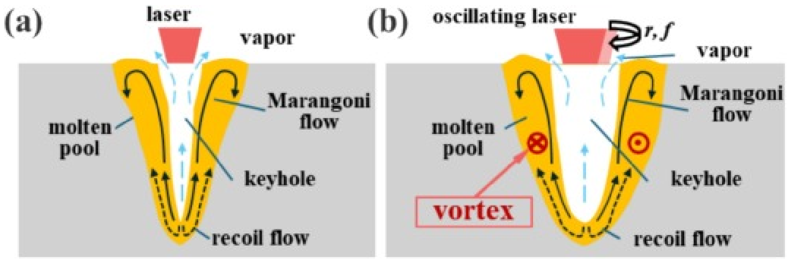

Fig. 10. Schematic diagram of flow in molten pool: (a) linear welding; (b) LBOW.

During the LBOW of the “sandwich” model, the laser beam oscillates across both sides of the metal-glass interface (Fig. 1 d). As the laser beam moves from point A to C along the glass side, it has a higher velocity. Conversely, the beam velocity is slower when moving from point C to A along the metal side. Therefore, the keyhole morphology keeps changing at the metal-glass interface. However, in the actual LBOW of Ta-10W, when the beam reaches point C, a portion of laser energy is used for forward penetration and melting of the solid base metal. As the beam moves backward to point A, the energy proportion acting on the rear molten pool gradually increases. This process causes further energy absorption in the molten pool, thereby increasing melting volumes and keyhole size [19]. Consequently, the “sandwich” model results consistently demonstrate that in LBOW, the keyhole depth at the metal-glass interface reaches its maximum at point A, while decreasing to its minimum at point C.

Fig. 11 shows the dynamic behavior of the keyhole in one cycle of parameters 1#, 3#, and 10#. Among them, Fig. 11 a and Fig. 11 b show the dynamic behavior of the keyhole under a small radius. Due to the minimum radius of 0.1 mm, the laser beam oscillation is close to the metal-glass interface, and the keyhole remains highly bright throughout the entire cycle. At f=50 Hz (1#), the overlap ratio is 0, resulting in a welding process that closely resembles linear welding. The keyhole morphology at the interface is spindle-shaped, with the rear wall exhibiting smoother characteristics than with no oscillation welding. However, gas bubbles entering the molten pool are observed (as shown in Fig. 11 a1). As the laser beam moves toward point C, the keyhole area decreases, accompanied by the gradual darkening of its rear and bottom regions. Conversely, the keyhole undergoes re-expansion as the laser moves toward point A. When the frequency increases to 300 Hz (3#), the keyhole morphology at point A exhibits a long columnar shape with a small opening size. Obvious protrusions and gas bubbles are observed along the rear keyhole wall. The keyhole depth gradually decreases during the move from point A to C. At the same time, its width expands, ultimately forming an inverted conical shape (Fig. 11 c). Similar to parameter 1#, the bottom region of the keyhole initially undergoes gradual darkening, followed by re-brightening during the beam return to point A. Fig. 11 c shows the dynamic behavior of the keyhole under a large radius and low frequency (10#). The beam energy distribution area becomes broader as the radius increases to 0.5 mm. When the laser beam reaches points B and D, it is far from the interface. Consequently, the variations in keyhole depth and brightness at the interface become more obvious. The maximum keyhole depth decreases while the inclination angle of the front wall increases, forming a wine glass shape that is wider at the top and narrower at the bottom. The slender bottom of the keyhole is prone to closing during the welding, resulting in bubble formation (Fig. 11 c1, c9). However, the overlap ratio and the molten pool length increase for parameter 10#, and the beam has a remelting effect on the rear wall of the keyhole. Therefore, small porosities are suppressed and only a few large porosities remain under this parameter.

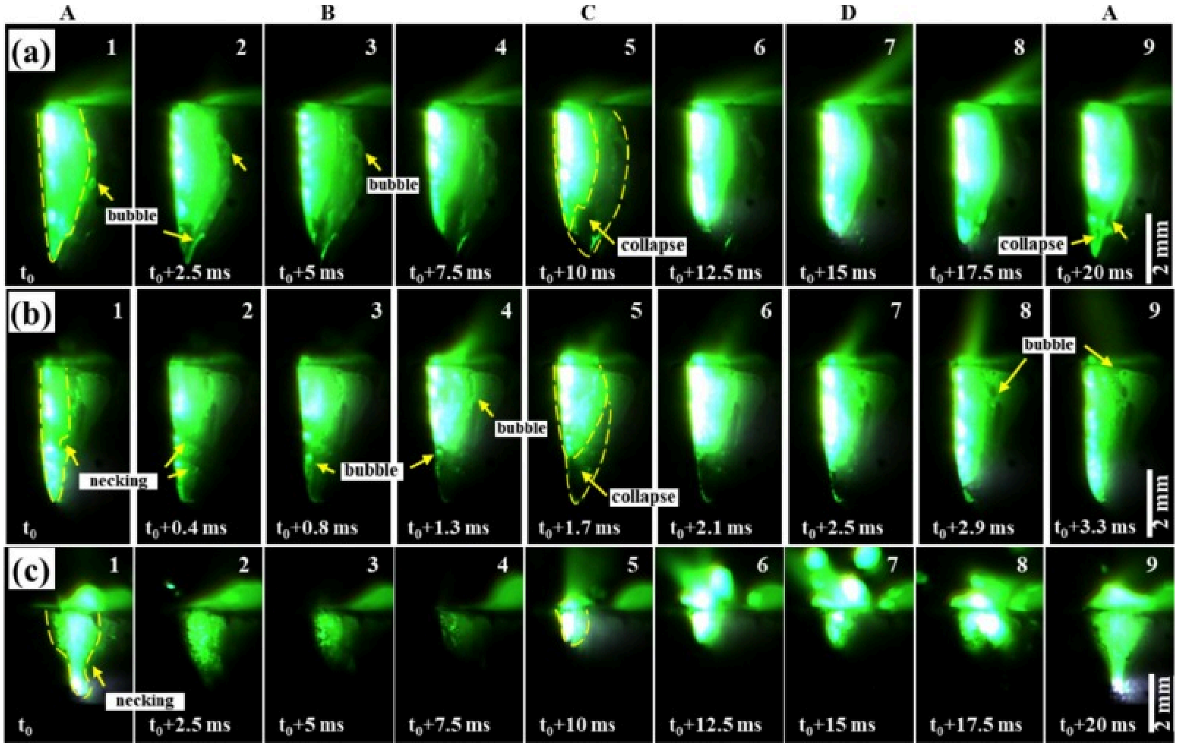

Fig. 11. Keyhole behavior of LBOW welds with high porosity: (a) 1#: r=0.1 mm, f=50 Hz; (b) 3#: r=0.1 mm, f=300 Hz; (c) 10#: r=0.5 mm, f=50 Hz.

Fig. 12 shows the dynamic behavior of the keyhole for parameters 9#, 11#, and 12#. The porosity was significantly suppressed under these parameters. Increasing oscillation frequency and radius decreases the peak beam energy density, reducing keyhole penetration depth and a wider fluctuation range of keyhole opening size. The keyhole is wide and shallow cup-shaped. At point A, both the bottom size of the keyhole and the inclination angle of its front wall exhibit significant increases compared to those under parameter 10#. Although the keyhole depth variation rule is similar to Fig. 11, the morphological changes are observed at the metal-glass interface: based on Fig. 12 a5, b5, c5. For the parameters with high oscillation frequency, as the beam moves to point C and forms a new shallower keyhole, the original deeper keyhole behind it keeps high brightness due to the delayed dissipation of metallic vapor. These two keyholes temporarily coexist under this condition. Immediately after that, the laser beam moves toward point A, causing the residual keyhole at the rear to expand again upon receiving laser energy. Concurrently, the shallow front keyhole merges with this rear keyhole, ultimately reforming a cup-shaped keyhole at the interface. For the parameters 9#, 11#, and 12#, the peak beam energy density decreases, accompanied by a reduced depth-to-width ratio of the keyhole, and the width of the keyhole bottom significantly expands. The stability of the keyhole increases, making it less likely to collapse and form gas holes. These reasons enhance the stability of the keyhole, reducing its susceptibility to collapse and thereby effectively suppressing porosity formation.

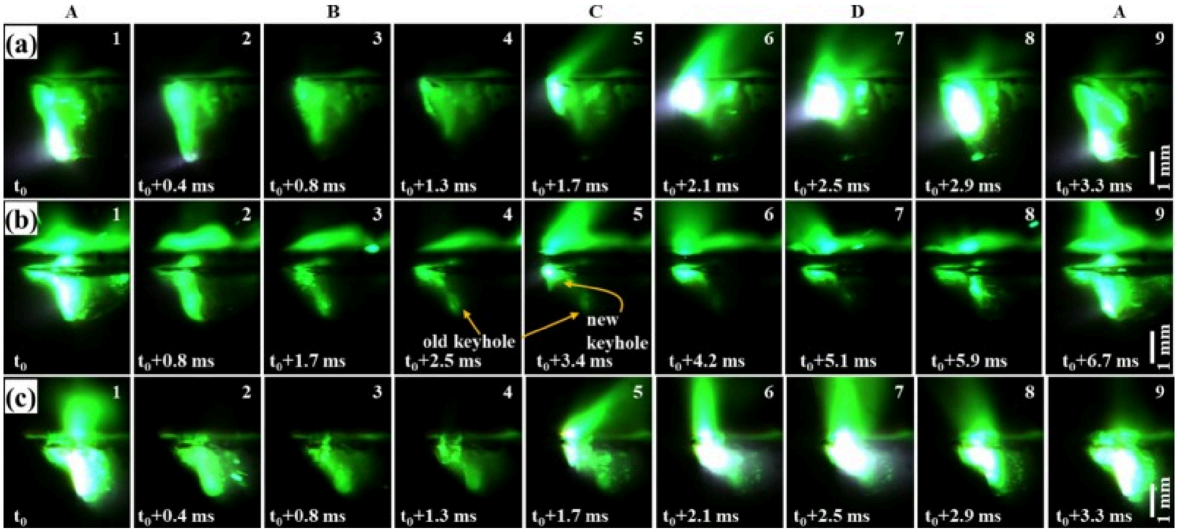

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Fig. 12. Keyhole behavior of LBOW welds with low porosity: (a) 9#: r=0.3 mm, f=300 Hz; (b) 11#: r=0.5 mm, f=150 Hz; (c) 12#: r=0.5 mm, f=300 Hz.

4. Analysis and discussion

In laser welding, keyhole-induced porosity primarily originates from the collapse of molten metal at the rear keyhole wall during welding. The application of circular beam oscillation enhances keyhole stability and modulates the molten metal flow around the keyhole and molten pool, thereby achieving effective porosity suppression. To further elucidate the mechanisms of porosity formation and suppression in LBOW of Ta-10W alloy, the schematics of keyhole and molten pool behaviors under various process conditions are presented in Fig. 13, based on the high-speed imaging observations and analyses in Section 3.3.

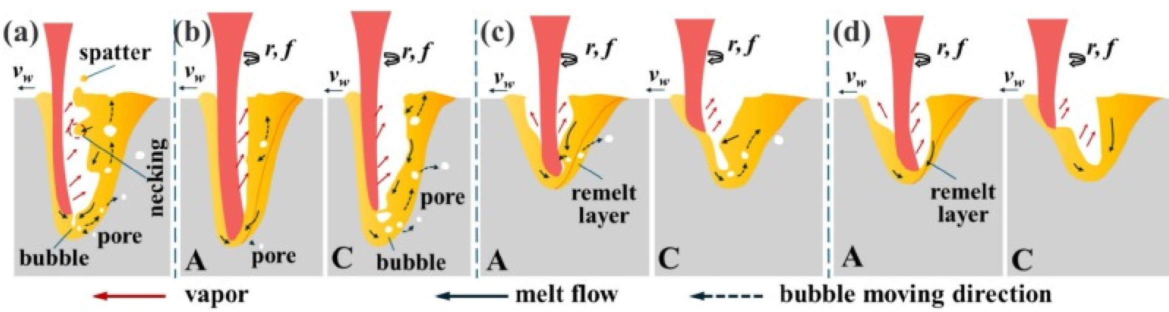

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Fig. 13. Mechanism of porosity generation and inhibition under different conditions: (a) no oscillation; (b) LBOW with small radius; (c) LBOW with large radius and low frequency; (d) LBOW with large radius and high frequency.

As shown in Fig. 13 a, in linear welding, the “necking” of the keyhole and the collapse of the rear wall directly led to the porosity formation. For LBOW, at small radii (Fig. 13 b), the darkening of the rear and bottom regions of the keyhole is due to the decrease in vapor pressure of the keyhole rear during forward laser beam motion. This leads to the gradual enlargement of protrusions on the keyhole rear wall, while molten metal fills the keyhole bottom under gravity effect (as shown in Fig. 11). This process may trap gas within the melt and cause bubble formation. The beam paths exhibit zero overlap ratio for parameter 1# with a 50 Hz frequency, resulting in negligible remelting effects on the molten pool during welding. Bubbles generated in previous cycles persist and fail to be eliminated, ultimately forming porosity. For parameter 3#, the increased path overlap ratio at 300 Hz enhances the remelting effect on the rear keyhole wall. During backward laser beam motion, this remelting promotes the absorption of bubbles within the molten pool, thereby significantly reducing porosity formation. However, the short molten pool and the narrow keyhole prevent pores from escaping. Furthermore, during keyhole fluctuations, the keyhole bottom is filled with molten metal and then re-deepened by laser beam irradiation in each oscillation cycle. Higher oscillation frequencies intensify the recurrence of this cyclic process, thereby substantially elevating the probability of bubble formation at the molten pool. Therefore, for parameter 3#, the weld porosity number within the reaches its maximum (Fig. 6 a).

When the radius increases to 0.5 mm, the molten pool length and the keyhole opening size expand. As shown in Fig. 13 c, under a large oscillation radius but with a frequency of 50 Hz, the path overlap ratio remains low, the keyhole moves along the beam oscillation path. The non-uniform energy distribution has a central low-energy zone, with path coincidence occurring only at points A and C. Although the slender keyhole bottom is prone to collapse-induced bubble formation, during the backward movement of the laser, the keyhole can also absorb bubbles in the molten pool while simultaneously reheating the molten metal or remelting the partially solidified rear molten pool. This promotes pores to rise and grow. Nonetheless, the velocity at which the bubbles rise is greater than the solidification velocity of the molten pool; as a result, weld 10# exhibits mainly sparse large pores. The path overlap ratio and laser beam velocity significantly increase with increasing frequency. When the overlap ratio exceeds 80%, and the laser beam velocity ranged from 437.90 to 975.82 mm/s (The range corresponding to 9#, 11#, 12#), porosity formation is effectively suppressed. As shown in Fig. 13 d, on the one hand, the combination of high beam velocity and path overlap ratio induces keyhole bottom widening, thereby enhancing keyhole resistance to collapse. On the other hand, the reheating and remelting effects on the rear molten pool during beam oscillation are enhanced, and the keyhole absorbs the bubbles in the molten pool more frequently during its movement. Therefore, under the oscillation parameters with large radius and high frequency, porosity is effectively eliminated within the LBOW welds [21,26,35].

According to the cross-sectional morphology of the LBOW welds, beam oscillation also has some influence on the porosity morphology. Fig. 14 provides magnified images of pores in selected welds under some parameters. In the no oscillation weld, the porosity exhibited nearly circular edges and smooth inner walls (Fig. 14 a). In LBOW welds, although some pores exhibit circular or oval shapes, their edge typically displays irregular and jagged shapes. Furthermore, the inner walls of the pore exhibit layered flow traces formed during molten metal solidification (Fig. 14 b, c). In addition to spherical porosities, slit-shaped or internal-filled porosities are observed in LBOW welds under some oscillation parameters, as shown in Fig. 14 d-f. This characteristic is more obvious in parameters with smaller radii or frequencies (such as parameters 1#~6#).

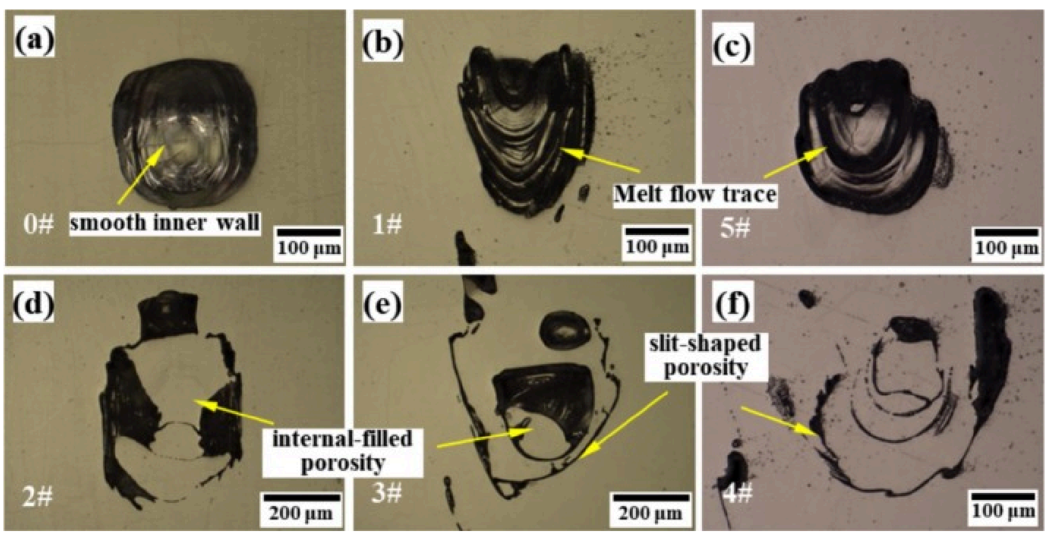

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Fig. 14. Enlarged view of weld pores: (a) circular pore of 0#; (b-c) circular pore of LBOW; (d-f) slit-shaped and internal-filled porosity of LBOW.

In the LBOW studies of aluminum alloys, steel and superalloys, laser beam oscillation virtually suppresses porosity in welds. Even if porosity is present in the welds, it usually appears as circular or oval porosities with relatively large sizes [17,36]. And no irregular porosities like those in Fig. 14 are observed.

This is mainly related to the special metal properties of the Ta-10W alloy. Fig. 15 shows the viscosity, density and surface tension of molten Ta, W, Al, Fe and Ni [[37], [38], [39]]. In the absence of reference data for viscosity and surface tension of molten Ta-10W alloy, quantitative comparisons are drawn between the viscosity (μ) and surface tension (σ) of Ta, W and those of Al, Fe, Ni. Compared with aluminum alloys, steel, and superalloys, the Ta-10W alloy has a higher melting point and greater density. It exhibits higher viscosity and surface tension in the molten state, resulting in inherently poor melt fluidity. According to Stokes' law, the bubble rising velocity (v) in a fluid is inversely proportional to the viscosity via the following equation:

$$v = \frac{gd^2(\rho_l - \rho_g)}{18\mu} \tag{6}$$

where g is the gravitational acceleration, d is the bubble diameter, and ρ_l and ρ_g represent the densities of the liquid and gas phases, respectively.

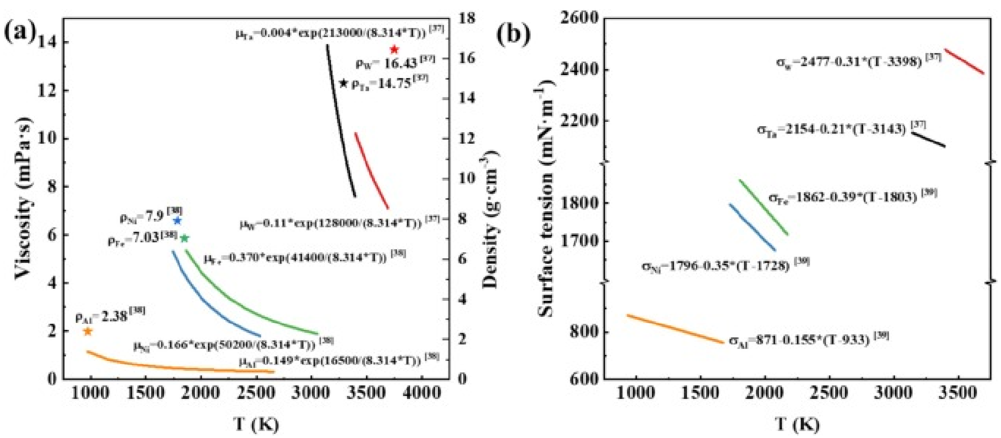

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Fig. 15. Material parameters of molten Ta, W, Al, Fe and Ni: (a) viscosity and density; (b) surface tension [[37], [38], [39].

Higher viscosity leads to slower bubble rising velocity in the molten metal [[40], [41], [42]]. Furthermore, by integrating the data from Fig. 15 and Eq.(6), we calculated the bubble rise velocity for quantitative analysis. For spherical bubbles with different diameters and assumed to be filled with air, the corresponding rise velocities in various molten materials are presented in Table 3. It can be observed from Table 3 that the bubble rising velocity in the Ta melt is the lowest, and its increase with pore size is also the most minimal. Given that the Ta-10W alloy consists of 90% Ta, its material properties are more closely aligned with those of pure tantalum. Therefore, compared with metals such as aluminum alloys and steels, Ta-10W alloy has a higher melting point, greater density, and a shorter molten pool during welding, accompanied by rapid solidification. The melt of Ta-10W exhibits higher viscosity and surface tension. This characteristic gives rise to a relatively slow bubble ascent velocity within the molten pool during welding. Bubbles have difficulty escaping before solidification, leading to severe porosity in Ta-10W alloy welds. In LBOW of Ta-10W alloy, the backward motion of the laser beam induces remelting of the molten pool rear wall and effectively suppresses porosity formation. However, the high melting point and poor fluidity of Ta-10W result in a thin remelted layer at the molten pool rear wall. When the overlap ratio falls below 75.6%, and the beam velocity decreases under 410.4mm/s, porosities in previously solidified welds are not fully eliminated. Instead, the pore walls near to the molten pool are partially opened, and the molten metal fills these partially open pores during beam oscillation. Being at a lower temperature, the filled melt fails to sufficiently remelt the pore walls and establish metallurgical bonding with them. This filled melt rapidly solidifies along the inner walls of pores, ultimately forming slit-shaped or internal-filled porosity morphologies in the LBOW welds.

Table 3. The rising velocities of bubbles with different diameters in molten Ta, W, Al, Fe and Ni.

	Bubble rising velocity v / mm· s^{-1}				
Bubble diameter d / mm	**Ta**	**W**	**Al**	**Fe**	**Ni**
0.1	5.80	8.77	11.279	7.18	8.12
0.5	144.90	219.25	281.98	179.37	203.09

Bubble diameter *d* / mm	Bubble rising velocity *v* / mm· s^{-1}				
	Ta	W	Al	Fe	Ni
1.0	562.81	877.02	1127.91	717.46	812.36

Additionally, the inhomogeneous energy distribution to circular beam oscillation and the resultant vortices in the molten pool may induce unilateral undercutting of the LBOW welds [30]. In this study, the density, viscosity, and surface tension of Ta-10W alloy are higher, so the molten metal of Ta-10W does not flow easily during welding when driven by the laser beam, and the unilateral undercut of LBOW welds is negligible. Slight unilateral undercutting on the weld surface occurs only under maximum radius and frequency, where beam velocity and molten pool vortex intensity reach peak values (weld 12# in Fig. 2).

5. Conclusions

This paper studied the laser beam oscillation welding (LBOW) of Ta-10W alloy. The performed analysis of weld morphology, porosity, keyhole, and molten pool dynamic behavior made it possible to draw the following conclusions:

(1) The weld formation and porosity suppression rules in LBOW of Ta-10W alloy were established. Under the oscillation parameters of r=0.3 mm/f=300 Hz and r=0.5 mm/f=150 Hz, the cross-section of the welds had a deep-penetration welding morphology. The weld surface showed good forming quality; the spatter was reduced, the molten pool and keyhole stability was improved. The porosity was reduced from 9% (0#) to 0.39% (9#) and 0.20% (11#).

(2) Using high-speed imaging and the "sandwich" model to analyze keyhole dynamics, the generation and suppression mechanisms of porosity during LBOW of Ta-10W alloy were systematically investigated. The velocity and path overlap ratio also increased as the oscillation radius and frequency increased. Under large radius and high frequency conditions, the opening and bottom sizes of the keyhole increased, and the keyhole had more opportunities to absorb bubbles in the molten pool during laser oscillation. The weld porosity was significantly suppressed when the beam velocity ranged from 437.90 to 975.82 mm/s and the path overlap ratio exceeded 80%.

(3) The Ta-10W alloy had a high melting point and density, with higher surface tension and viscosity in the molten state. This resulted in poor fluidity of molten melt during welding and slow bubble rise velocity in the molten pool. At the path overlap ratio below 75.6% and the beam velocity below 410.4mm/s, the insufficient beam remelting efficacy led to the formation of slit-shaped and internal-filled porosity within the LBOW welds.

CRediT authorship contribution statement

Xin Du: Writing – original draft, Visualization, Validation, Investigation, Data curation, Formal analysis. **Qiang Wu:** Writing – review & editing, Resources, Project administration, Funding acquisition, Conceptualization, Methodology. **Zekang Chen:** Data curation, Validation. **Chengbing Yao:** Formal analysis, Validation. **Jianglin Zou:** Visualization, Validation. **Rongshi Xiao:** Resources, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Recommended articles

Data availability

Data will be made available on request.

References

[1] J.H. Perepezko
Materials science. The hotter the engine, the better
Science, 326 (5956) (2009), pp. 1068-1069, 10.1126/science.1179327
View in Scopus Google Scholar

[2] Y.J. Yu, Y. Xu, X.Y. Duan, S. Zhu, Y.P. Lei, H.Q. Xia, A. Wang, Q. Jiang, J.C. Tang
Process optimization and performance of spark plasma sintered Ta-10W alloy
Intermetallics, 175 (2024), Article 108495, 10.1016/j.intermet.2024.108495
View PDF View article View in Scopus Google Scholar

[3] S. Kim, T.G. Kim, C. Kim, S.J. Chang, D. Kim, S. Yang, H.U. Hong
Microstructure evolution and cyclic thermal behavior of ta-10W cladded onto the inner surface of steel tube by explosive welding
Mater. Sci. Eng. A, 923 (2025), Article 147716, 10.1016/j.msea.2024.147716
View PDF View article View in Scopus Google Scholar

[4] S.M.H. Hojjatzadeh, N.D. Parab, Q. Guo, M. Qu, L. Xiong, C. Zhao, L.I. Escano, K. Fezzaa, W. Everhart, T. Sun, L. Chen
Direct observation of pore formation mechanisms during LPBF additive manufacturing process and high energy density laser welding
Int J Mach Tool Manu, 153 (2020), Article 103555, 10.1016/j.ijmachtools.2020.103555
View PDF View article View in Scopus Google Scholar

[5] R. Cunningham, C. Zhao, N. Parab, C. Kantzos, J. Pauza, K. Fezzaa, T. Sun, A.D. Rollett
Keyhole threshold and morphology in laser melting revealed by ultrahigh-speed x-ray imaging
Science, 363 (2019), pp. 849-852, 10.1126/science.aav4687
View in Scopus Google Scholar

[6] X.M. Meng, M. Bachmann, F. Yang, M. Rethmeier
Toward prediction and insight of porosity formation in laser welding: a physics-informed deep learning framework
Acta Mater., 286 (2025), Article 120740, 10.1016/j.actamat.2025.120740
View PDF View article View in Scopus Google Scholar

[7] X. Du, B.X. Qi, Z.J. Zhao, T. Zhang, Q. Wu, J.L. Zou, R.S. Xiao
Process research on fiber laser welding of Ta-10W refractory alloy
Chin. J. Lasers, 49 (16) (2022), Article 1602006, 10.3788/CJL202249.1602006
View in Scopus Google Scholar

[8] J.F. Gong, Y.C. Huang, M.X. Yang, L.Q. Li, Z.H. Zhang, P. He
Numerical simulation analysis of pore suppression behavior in vacuum laser lap welding of Ta10W alloy
Opt. Laser Technol., 183 (2025), Article 112403, 10.1016/j.optlastec.2024.112403
View PDF View article View in Scopus Google Scholar

[9] C. Hagenlocher, M. Sommer, F. Fetzer, R. Weber, T. Graf
Optimization of the solidification conditions by means of beam oscillation during laser beam welding of aluminum

Mater. Des., 160 (2018), pp. 1178-1185, 10.1016/j.matdes.2018.11.009 ↗

View PDF View article View in Scopus ↗ Google Scholar ↗

[10] C. Zhang, Y. Yu, C. Chen, X. Zeng, M. Gao
Suppressing porosity of a laser keyhole welded Al-6Mg alloy via beam oscillation
J. Mater. Process. Technol., 278 (2020), Article 116382, 10.1016/j.jmatprotec.2019.116382 ↗

View PDF View article View in Scopus ↗ Google Scholar ↗

[11] H. Sebestova, M. Jambor, P. Hornik, J. Novotny, L. Mrna
Laser beam oscillation welding for fatigue properties enhancement of tailor-welded blanks
Thin-Walled Struct., 196 (2024), Article 111506, 10.1016/j.tws.2023.111506 ↗

View PDF View article View in Scopus ↗ Google Scholar ↗

[12] P. Horník, H. Šebestová, J. Novotný, L. Mrňa
Laser beam oscillation strategy for weld geometry variation
J. Manuf. Process., 84 (2022), pp. 216-222, 10.1016/j.jmapro.2022.10.016 ↗

View PDF View article View in Scopus ↗ Google Scholar ↗

[13] T. Li, D. Zhou, Y. Yan, J. Liu
Effect of beam oscillation on intermetallic compounds and mechanical properties of steel/aluminum laser welded joint
J. Manuf. Process., 73 (2022), pp. 40-53, 10.1016/j.jmapro.2021.10.066 ↗

View PDF View article View in Scopus ↗ Google Scholar ↗

[14] X.B. Pang, J.H. Dai, S. Chen, M.J. Zhang
Microstructure and mechanical properties of fiber laser welding of aluminum alloy with beam oscillation
Appl. Sci., 9 (23) (2019), p. 5096, 10.3390/app9235096 ↗

View in Scopus ↗ Google Scholar ↗

[15] T. Girerd, A. Gameros, M. Simonelli, A. Norton, A.T. Clare
Modulation of melt pool behaviour using novel laser beam oscillation methods
J. Mater. Process. Technol., 325 (2024), Article 118300, 10.1016/j.jmatprotec.2024.118300 ↗

View PDF View article View in Scopus ↗ Google Scholar ↗

[16] D. Franco, J.P. Oliveira, T.G. Santos, R.M. Miranda
Analysis of copper sheets welded by fiber laser with beam oscillation
Opt. Laser Technol., 133 (2021), Article 106563, 10.1016/j.optlastec.2020.106563 ↗

View PDF View article View in Scopus ↗ Google Scholar ↗

[17] T.S. Kabasakaloglu, M. Erdogan
Characterisation of figure-eight shaped oscillation laser welding behaviour of 5083 aluminium alloy
Sci. Technol. Weld. Join., 25 (7) (2020), pp. 609-616, 10.1080/13621718.2020.1794652 ↗

View in Scopus ↗ Google Scholar ↗

[18] G. Chen, B. Wang, S. Mao, P. Zhong, J. He
Research on the “∞”-shaped laser scanning welding process for aluminum alloy
Opt. Laser Technol., 115 (2019), pp. 32-41, 10.1016/j.optlastec.2019.01.046 ↗

View PDF View article View in Scopus ↗ Google Scholar ↗

[19] X. Chen, N. Jiang, M. Jiang, Y. Du, S.C. Ma, Y. Chen, C.W. Tan, Z.L. Lei, S.C. Zhao, Y.B. Chen
Numerical investigation of asymmetric weld fusion geometry in laser welding of aluminium alloy with beam oscillation
Sci. Technol. Weld. Join., 27 (8) (2022), pp. 595-605, 10.1080/13621718.2022.2091343 ↗

Google Scholar ↗

[20] A. Sadeghian, S. Nath, Y.Z. Huang, R.S. Matharu, N. Wadee, N. Pembrey, D.G. Waugh
Quasi-continuous wave pulsed laser welding of copper lap joints using spatial beam oscillation
Micromachines, 13 (12) (2022), p. 2092, 10.3390/mi13122092 ↗
View in Scopus ↗ Google Scholar ↗

[21] L. Shi, X. Li, L. Jiang, M. Gao
Numerical study of keyhole-induced porosity suppression mechanism in laser welding with beam oscillation
Sci. Technol. Weld. Join., 26 (5) (2021), pp. 349-355, 10.1080/13621718.2021.1913562 ↗
View in Scopus ↗ Google Scholar ↗

[22] F. Fetzer, M. Sommer, R. Weber, J.P. Weberpals, T. Graf
Reduction of pores by means of laser beam oscillation during remote welding of AlMgSi
Opt. Lasers Eng., 108 (2018), pp. 68-77, 10.1016/j.optlaseng.2018.04.012 ↗
View PDF View article View in Scopus ↗ Google Scholar ↗

[23] S. Börner, D. Dittrich, T. Sahm, L. Ullmann, P.H. Kamm, A. Jahn
X-ray analysis of capillary and melt pool dynamics using laser beam oscillation welding for aluminum die-casts
J. Laser Appl., 37 (2) (2025), Article 022019, 10.2351/7.0001639 ↗
View in Scopus ↗ Google Scholar ↗

[24] A. Mohan, D. Ceglarek, P. Franciosa, M. Auinger
Numerical study of beam oscillation and its effect on the solidification parameters and grain morphology in remote laser welding of high-strength aluminium alloys
Sci. Technol. Weld. Join., 28 (5) (2023), pp. 362-371, 10.1080/13621718.2022.2163341 ↗
View in Scopus ↗ Google Scholar ↗

[25] A. Mohan, P. Franciosa, D. Ceglarek, M. Auinger
Numerical simulation of transport phenomena and its effect on the weld profile and solute distribution during laser welding of dissimilar aluminium alloys with and without beam oscillation
Int. J. Adv. Manuf. Technol., 124 (10) (2023), pp. 3311-3325, 10.1007/s00170-022-10623-3 ↗
View in Scopus ↗ Google Scholar ↗

[26] M.P. Wu, Z. Luo, Y. Li, L. Liu, S. Ao
Effect of oscillation modes on weld formation and pores of laser welding in the horizontal position
Opt. Laser Technol., 158 (2023), Article 108801, 10.1016/j.optlastec.2022.108801 ↗
View PDF View article View in Scopus ↗ Google Scholar ↗

[27] H. Yang, G. Jing, P. Gao, Z. Wang, X. Li
Effects of circular beam oscillation technique on formability and solidification behaviour of selective laser melted Inconel 718: from single tracks to cuboid samples
J. Mater. Sci. Technol., 51 (2020), pp. 137-150, 10.1016/j.jmst.2019.09.044 ↗
View PDF View article Google Scholar ↗

[28] ASM International
Properties and Selection: Nonferrous Alloys and Special-Purpose Materials
vol. 2, ASM International (1990)
Google Scholar ↗

[29] N. Cao
The Study on Working Head for Titanium Laser Welding
Thesis
Beijing University of Technology, Beijing (2006)
Google Scholar ↗

[30] S.G. Kang, J. Shin
Laser beam oscillation welding of aluminum alloy using the spatially modulated beam by diffractive optical element (DOE)
J. Manuf. Process., 66 (2021), pp. 387-396, 10.1016/j.jmapro.2021.04.029
View PDF View article View in Scopus Google Scholar

[31] G. Chen, Q. Yin, X. Shu, Y. Bi, B. Zhang, J. Feng
Microstructure and properties of electron beam welded joints of tantalum and tungsten
Weld World, 62 (2018), pp. 775-782, 10.1007/s40194-018-0600-z
View in Scopus Google Scholar

[32] M. Gao, H. Wang, K. Hao, H. Mu, X. Zeng
Evolutions in microstructure and mechanical properties of laser lap welded AZ31 magnesium alloy via beam oscillation
J. Manuf. Process., 45 (2019), pp. 92-99, 10.1016/j.jmapro.2019.07.001
View PDF View article View in Scopus Google Scholar

[33] Z.S. Ren, L. Gao, S.J. Clarke, K. Fezzaa, P. Shevchenko, A. Choi, *et al.*
Machine learning–aided real-time detection of keyhole pore generation in laser powder bed fusion
Science, 379 (6627) (2023), pp. 89-94, 10.1126/science.add4667
View in Scopus Google Scholar

[34] W. Huang, W. Cai, T.J. Rinker, J. Bracey, W. Tan
Effects of laser oscillation on metal mixing, microstructure, and mechanical property of aluminum–copper welds
Int J Mach Tool Manu, 188 (2023), Article 104020, 10.1016/j.ijmachtools.2023.104020
View PDF View article View in Scopus Google Scholar

[35] J. Li, Y. Liu, Y. Tao, Q. Zhang, K. Kang, Q. Sun, *et al.*
Analysis of vapor plume and keyhole dynamics in laser welding stainless steel with beam oscillation
Infrared Phys. Technol., 113 (2021), Article 103536, 10.1016/j.infrared.2020.103536
View PDF View article View in Scopus Google Scholar

[36] P. Xia, L. Xiong, Y. Hu, X. Feng, C. Wang
Effect of laser oscillation and beam incident angle on porosity in double-sided filler welding of 2219 aluminum alloy T joint
Opt. Laser Technol., 166 (2023), Article 109581, 10.1016/j.optlastec.2023.109581
View PDF View article View in Scopus Google Scholar

[37] T. Ishikawa, P.F. Paradis
Thermophysical properties of molten refractory metals measured by an electrostatic levitator
J. Electron. Mater., 34 (2005), pp. 1526-1532, 10.1007/s11664-005-0160-z
View in Scopus Google Scholar

[38] T. Dinsdale, P.N. Quested
The viscosity of aluminium and its alloys-a review of data and models
J. Mater. Sci., 39 (2004), pp. 7221-7228, 10.1023/B:JMSC.0000048735.50256.96
View in Scopus Google Scholar

[39] Z.F. Yuan, J.J. Ke
Surface Tension of Metals and Alloys
Science Press, Beijing (2006)
(In Chinese)
Google Scholar

[40] S. Li, L.Q. Sun, A.M. Zhang
Dynamic behavior of rising bubble
Acta Phys. Sin., 63 (18) (2014), Article 184701, 10.7498/aps.63.184701
View in Scopus Google Scholar

[41] A.M. Zhang, S.M. Li, S. Li, Y.L. Liu
Advancements in bubble dynamics research
Chin. J. Theor. Appl. Mech., 57 (1) (2025), pp. 1-16
Google Scholar

[42] M. Iguchi, A. Kawajiri, H. Tomida, Z. Morita
Cold model experiments on the dynamic behavior of bubbles rising in highly viscous-liquid
J. Iron Steel Inst. Jpn., 78 (9) (1992), pp. 1456-1463, 10.2355/tetsutohagane1955.78.9_1456
Google Scholar

Cited by (1)

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